X1 Capstone 1: refactor a messy pipeline

Refactor a notebook-style pipeline to the standard, with tests on a local SparkSession.

X1-L1 Capstone 1: refactor the NAT pipeline

Refactor one function at a time, in pipeline order, with acceptance tests running after every change.

Capstone 1 workflow	Read and run the acceptance tests, then refactor one function at a time in pipeline order.
Where the capstone files live	Start: practice/capstone/before_pipeline.py. Yours: practice/exercises/x1_pipeline.py. Tests: practice/tests/test_x1_pipeline.py.
When to open the reference solution	Only after your tests pass. Then compare structure, not just results.